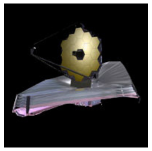

9624 - Constraining Collisions in the beta Pic Terrestrial Planet Zone

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
beta-Pic_MRS				
	1	beta Pic Background - MIRI MRS Spectroscopy	MIRI Medium Resolution Spectroscopy	(2) -BET-PIC
	2	beta Pic star - MIRI MRS Spectroscopy	MIRI Medium Resolution Spectroscopy	(2) -BET-PIC
	3	beta Pic PSF - MIRI MRS Spectroscopy	MIRI Medium Resolution Spectroscopy	(3) -N-CAR

ABSTRACT

The beta Pic planetary system is dynamically active with on-going collisions among planetesimals, releasing gas and fine dust. In addition, JWST MIRI observations have recently revealed dramatic changes in the mid-infrared continuum and silicate emission features arising from the debris disk during the past 20 years. These observations indicate that there was a massive asteroid collision in the habitable zone just prior to the Spitzer observations in 2004-5. We propose to obtain MIRI MRS observations in Cycle 5 to better understand (1) the evolution of the dust spectra (shockingly different between Spitzer and JWST epochs) or underlying steady state behavior; (2) the evolution of the atomic gas that results from the on-going collisional destruction of volatile-rich planetesimals; (3) the on-going collisional production of sub-blow out size particles responsible for the dusty wind in the beta Pic disk; (4) the duty-cycle of large collisions, known to be frequent in the system, and how they might be connected to the orbital dynamics of the planetary system.

OBSERVING DESCRIPTION

We propose to obtain multi-epoch MIRI/MRS observations of beta Pic to characterize the time variability of its debris disk and constrain the location of collisions within the disk. In addition to the observations of beta Pic, we also request observations of (1) a nearby background field 10” from beta Pictoris to enable empirical background subtraction (as recommended by STScI) and a (2) PSF calibrator star to enable empirical PSF subtraction to characterize the disk emission close to the star.

Thus, for each epoch, we request the following sequence with no interrupts:

1. Background Sky Observation
2. Beta Pic MIRI/MRS Observation
3. HR 2435 MIRI/MRS Calibrator Observation

For both the science target and the PSF calibrator, we request one mosaic field (to focus on characterizing dust near the bright host star, including beta Pic b) and the same dither pattern (4-Point Dither Pattern for Extended Sources in the Negative direction). We choose the Number of Groups per Integration to avoid saturation on the bright star and total exposure times to yield the same PSF SNR. Finally, we request Target Acquisition to ensure that the source is well centered within an MRS slice in addition to within the MRS Field-of-View.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	-BET-PIC	RA: 05 47 17.0953 (86.8212304d) Dec: -51 03 58.15 (-51.06615d) Equinox: J2000	Proper Motion RA: 4.932985550191305E-4 sec of time/yr Proper Motion Dec: 0.0831 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Circumstellar disks, Circumstellar dust, Circumstellar gas, Debris disks] Extended=YES				
(3)	-N-CAR	RA: 06 34 58.5668 (98.7440283d) Dec: -52 58 32.03 (-52.97556d) Equinox: J2000	Proper Motion RA: -8.469571867308019E-4 sec of time/yr Proper Motion Dec: 0.010539999999999999 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Calibration Description=[A stars, Point spread function]				

Observation

Proposal 9624, Observation 1: beta Pic Background - MIRI MRS Spectroscopy

Wed Jun 03 15:00:19 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(beta Pic Background - MIRI MRS Spectroscopy (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	-BET-PIC	RA: 05 47 17.0953 (86.8212304d) Dec: -51 03 58.15 (-51.06615d) Equinox: J2000	Proper Motion RA: 4.932985550191305E-4 sec of time/yr Proper Motion Dec: 0.0831 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Circumstellar disks, Circumstellar dust, Circumstellar gas, Debris disks]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	12	8	1	Dither 1	4	32	1143.316	
1	SHORT(A)	MRSLONG		FASTR1	15	7	1	Dither 1	4	28	1232.118	
1	SHORT(A)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717	
2		IMAGER	F1500W	FASTR1	9	11	1	Dither 1	4	44	1209.917	
2	MEDIUM(B)	MRSLONG		FASTR1	15	7	1	Dither 1	4	28	1232.118	
2	MEDIUM(B)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717	
3		IMAGER	F2550W	FASTR1	8	12	1	Dither 1	4	48	1187.717	
3	LONG(C)	MRSLONG		FASTR1	15	7	1	Dither 1	4	28	1232.118	
3	LONG(C)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717	

Special Requirements	
	Sequence Observations 1, 2, 3, Non-interruptible Same V3 PA 1, 2 (Aperture PAs differ)

Observation

Proposal 9624, Observation 2: beta Pic star - MIRI MRS Spectroscopy Wed Jun 03 15:00:19 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(beta Pic star - MIRI MRS Spectroscopy (Obs 2)) Warning (Form): Imager Filter overlap.

(beta Pic star - MIRI MRS Spectroscopy (Obs 2)) Warning (Form): Imager Filter overlap.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(beta Pic star - MIRI MRS Spectroscopy (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	-BET-PIC	RA: 05 47 17.0953 (86.8212304d) Dec: -51 03 58.15 (-51.06615d) Equinox: J2000	Proper Motion RA: 4.932985550191305E-4 sec of time/yr Proper Motion Dec: 0.0831 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Circumstellar disks, Circumstellar dust, Circumstellar gas, Debris disks]
Extended=YES

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	FND	FAST	4	1	1	11.1	218192.1

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	12	8	1	Dither 1	4	32	1143.316	
1	SHORT(A)	MRSLONG		FASTR1	15	7	1	Dither 1	4	28	1232.118	
1	SHORT(A)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717	
2		IMAGER	F1500W	FASTR1	9	11	1	Dither 1	4	44	1209.917	
2	MEDIUM(B)	MRSLONG		FASTR1	15	7	1	Dither 1	4	28	1232.118	
2	MEDIUM(B)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717	
3		IMAGER	F2550W	FASTR1	8	12	1	Dither 1	4	48	1187.717	
3	LONG(C)	MRSLONG		FASTR1	15	7	1	Dither 1	4	28	1232.118	
3	LONG(C)	MRSSHORT		FASTR1	8	12	1	Dither 1	4	48	1187.717	

Special Requirements	Aperture PA Range 16 to 32 Degrees (V3 16.0 to 32.0) Sequence Observations 1, 2, 3, Non-interruptible Same V3 PA 1, 2 (Aperture PAs differ)

Observation

Proposal 9624, Observation 3: beta Pic PSF - MIRI MRS Spectroscopy

Wed Jun 03 15:00:19 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	-N-CAR	RA: 06 34 58.5668 (98.7440283d) Dec: -52 58 32.03 (-52.97556d) Equinox: J2000	Proper Motion RA: -8.469571867308019E-4 sec of time/yr Proper Motion Dec: 0.010539999999999999 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Calibration
Description=[A stars, Point spread function]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	FND	FAST	4	1	1	11.1	133000.13

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	SHORT(A)	MRSLONG		FASTR1	30	7	1	Dither 1	4	28	2397.635	
1	SHORT(A)	MRSSHORT		FASTR1	16	12	1	Dither 1	4	48	2253.332	
2	MEDIUM(B)	MRSLONG		FASTR1	30	7	1	Dither 1	4	28	2397.635	
2	MEDIUM(B)	MRSSHORT		FASTR1	16	12	1	Dither 1	4	48	2253.332	
3	LONG(C)	MRSLONG		FASTR1	30	7	1	Dither 1	4	28	2397.635	
3	LONG(C)	MRSSHORT		FASTR1	16	12	1	Dither 1	4	48	2253.332	

Special Requirements	Sequence Observations 1, 2, 3, Non-interruptible